Robustness for SISO Systems

What a feedback loop can promise

A controller does not act on the true plant model. It acts on an error signal. The model is our tool for predicting whether that action corrects the error or amplifies it. Robust control asks how wrong the prediction may be before stability or performance is lost.

> Read sections 1–8 as one argument: signals → hidden modes → Nyquist geometry → margins → model sets → small gain → performance → a delayed satellite. Section 9 is a separate worked guide to all ten exercises on book pp. 42–45. Background: rational transfer functions, eigenvalues, and basic complex analysis.

1 Derive the signals before naming the functions

Use **negative feedback**, with an input disturbance d and measurement noise n:

$$e = r - y - n, \quad u = Ke + d, \quad y = Gu. \tag{1}$$

Here u is the total plant input, not just the controller output. Solving the equations,

$$(1 + GK)e = r - Gd - n. \tag{2}$$

Define the loop gain $L = GK$, sensitivity $S = (1 + L)^{-1}$, and complementary sensitivity $T = L(1 + L)^{-1}$. Then

$$e = Sr - GSd - Sn, \quad u = KSr + Sd - KSn, \tag{3}$$

$$y = Tr + GSd - Tn, \quad S + T = 1. \tag{4}$$

These identities explain what each function measures. Small S gives good reference tracking and rejection of output disturbances; small GS rejects plant-input disturbances. Small T limits transmission of measurement noise to the physical output. Small KS limits the command demanded by references and noise. A low tracking error alone says nothing about actuator saturation.

For a reference sinusoid at frequency ω, the steady-state error amplitude is $|S(i\omega)|$ times its amplitude. Thus high loop gain usually helps tracking: $|L|$ large gives $S \approx \frac{1}{L}$ and $T \approx 1$. At high frequency, where sensing is noisy and the model is poor, one instead wants T small. Since $S + T = 1$, they cannot both be small at the same frequency. This is an algebraic conflict, before any optimization.

All rational models below are real and proper. “Stable” means proper with poles strictly in the left half-plane. Well-posedness additionally requires $1 + G(\infty)K(\infty) \neq 0$: otherwise the instantaneous feedback equations cannot be solved causally. Zero initial conditions are implicit in transfer equations.

2 Stability must include the signals you did not plot

Set $n = 0$ in Equation 1. The two equations at the summing junctions give

$$\begin{pmatrix} 1 & G \\ -K & 1 \end{pmatrix} \begin{pmatrix} e \\ u \end{pmatrix} = \begin{pmatrix} r \\ d \end{pmatrix}, \quad \begin{pmatrix} e \\ u \end{pmatrix} = \begin{pmatrix} S & -GS \\ KS & S \end{pmatrix} \begin{pmatrix} r \\ d \end{pmatrix}. \tag{5}$$

Internal stability of the transfer interconnection means that every entry of this inverse is stable: S, GS, and KS. Then $T = 1 - S$ is stable too. Stability of S alone is only stability of one observed input–output channel.

For minimal plant and controller realizations, this is equivalent to the autonomous closed-loop state matrix being Hurwitz. For arbitrary nonminimal realizations, hidden unstable states must be checked separately: no transfer test can see a state that is unreachable and unobservable even before feedback.

2.1 Why an unstable cancellation is not a cure

Let $G = \frac{1}{s-1}$ and $K = \frac{s-1}{s+2}$. The product reduces to $L = \frac{1}{s+2}$, giving

$$S = \frac{s+2}{s+3}, \quad KS = \frac{s-1}{s+3}, \quad GS = \frac{s+2}{(s-1)(s+3)}. \tag{6}$$

The reference-to-error plot looks stable. But an input disturbance still excites the plant's growing mode at $s = 1$. The zero of the controller prevented that mode from appearing in the reduced product; it did not remove the plant state. Cancelling a stable pole is not the same danger, although approximate cancellation can still harm transients.

For a general check, write $G = \frac{N_G}{D_G}$, $K = \frac{N_K}{D_K}$, each pair coprime, and retain the **unreduced interconnection polynomial**

$$p = D_G D_K + N_G N_K. \tag{7}$$

Then

$$\begin{pmatrix} S & -GS \\ KS & S \end{pmatrix} = \frac{1}{p} \begin{pmatrix} D_G D_K & -N_G D_K \\ D_G N_K & D_G D_K \end{pmatrix}. \tag{8}$$

If p is Hurwitz and the loop is well posed, every entry is stable. Conversely, if $p(\lambda) = 0$ in the closed right half-plane, coprimeness prevents all three distinct numerators from vanishing there: at least one entry exposes that bad root.

This yields the useful version of book Theorem 2.24: internal stability is equivalent to **both** stability of S **and** absence of closed-right-half-plane pole–zero cancellation when forming GK. Absence of cancellation by itself does not stabilize anything.

> Check the unreduced states or all internal transfer channels first. Frequency robustness tests will assume this nominal internal stability; they cannot repair its absence.

3 Nyquist counts unstable modes without solving for them

Why should a curve around -1 know the roots of a characteristic polynomial? Because the closed-loop poles are zeros of $1 + L$, while the open-loop poles are its poles. The argument principle converts their difference into a winding number.

Choose the book's contour Γ: it surrounds the right half-plane **clockwise**, with small detours into the left half-plane around imaginary-axis poles so those poles remain inside. It avoids all poles. Let P count the enclosed poles of L, Z the zeros of $1 + L$, and N the net counterclockwise turns of $L(\Gamma)$ around -1. Then

$$N = \frac{1}{2\pi}\Delta_\Gamma \arg(1 + L) = P - Z. \tag{9}$$

The clockwise orientation of the **domain** supplies the minus sign relative to the usual positively oriented argument principle. Translating the image curve by 1 does not reverse its orientation. A counterclockwise turn remains counterclockwise.

Provided the curve avoids -1, $L(\infty) \neq -1$, and the outer radius contains all relevant roots, S is stable exactly when $N = P$. For rational L the radius may be taken to infinity. If GK has no unstable cancellations, P is the sum of the unstable/marginal pole counts of G and K, and this becomes an internal-stability test.

3.1 A one-pole example you can solve both ways

Take an unstable scalar process $\dot{x} = x + u$, $y = x$, and proportional feedback $u = k(r - y)$. Then $L = \frac{k}{s-1}$ and the autonomous pole is $1 - k$. For $k > 1$ it is stable. The image of the imaginary axis is the circle

$$\left(\Re L + \frac{k}{2}\right)^2 + (\Im L)^2 = \left(\frac{k}{2}\right)^2. \tag{10}$$

It passes through 0 and $-k$. With the clockwise right-half-plane contour the circle is traversed counterclockwise. It encloses -1 exactly for $k > 1$, giving $N = 1 = P$. At $k = 1$ the curve passes through -1, which is the closed-loop pole at zero. At $0 < k < 1$ there are no encirclements, so $Z = P - N = 1$: the feedback was too weak.

This example also explains why "no encirclement of -1" is not a universal stability rule. It is correct only when the open loop has no unstable poles to compensate. For an integrator or a flexible satellite, the small contour detours matter; a cropped plot of $L(i\omega)$ alone omits them. Count a **closed, oriented contour**, not an unoriented sketch.

In robust analysis we continuously deform this curve. If it cannot touch -1 and its relevant pole count does not change, its winding number cannot change either. This topological observation is the bridge from classical margins to the weighted test.

4 Classical margins are restricted directions of model error

A real gain change $L \mapsto kL$, $k > 0$, stretches the Nyquist curve radially. At a negative-real-axis crossing, loss of stability can occur at $k = -\frac{1}{L(i\omega)}$. The gain margins are the endpoints of the stable interval containing $k = 1$; enumerate all crossings and check which intervals are stable, including possible endpoints at infinity.

A constant phase change rotates the curve. Only points with $|L| = 1$ can then hit -1. The symmetric phase margin considers rotation in **both** directions. It need not equal the tolerance to a physical delay, which adds lag $-\omega\tau$ rather than a constant phase. For a nominally stable, strictly proper loop, list every positive unit-gain crossing and form

$$\tau_j = \frac{\left(\arg L(i\omega_j) + \pi\right) \bmod (2\pi)}{\omega_j}. \tag{11}$$

The first positive candidate is the first delay stability boundary, subject to checking degenerate crossings. Angles are in radians. A single $\frac{\varphi_m}{\omega_c}$ is valid only when that crossing really supplies the earliest lag-induced encounter.

The satellite in section 8 has **three** crossings: $\omega = 0.4378, 0.9407, 1.0879$ rad/s. Their positive-delay candidates are $1.2664, 5.3456, 0.7669$ s. Choosing the first crossing would overstate the delay tolerance by 65%. A negative signed angular clearance at the middle crossing does not mean this nominal loop is unstable.

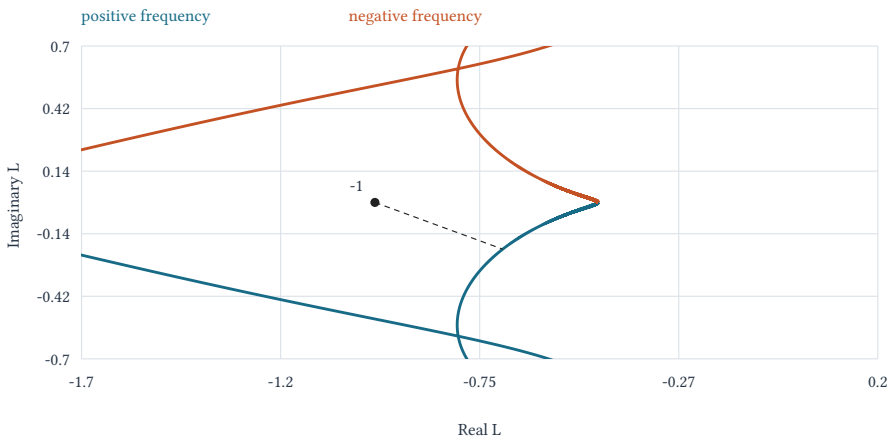


Figure 1: Book Example 2.10, a zoom of the positive/negative-frequency traces. The dangerous point is -1 on the real axis; this plot is geometry, not a complete Nyquist contour or a pole-count certificate.

Mixed gain and phase changes can pass between the directions tested separately. For the book's fifth-order loop (coefficients in `scripts/figures.m`), a frequency search finds $\min|1 + L| \approx 0.36976$ near 3.0574 rad/s. The complex multiplier $c \approx 1.31921 - 0.39697i$ makes $cL = -1$ there: gain 1.37765 and phase $-16.747°$ act **together**. The nominal characteristic roots have maximum real part -0.15904. Applied separately, that gain and that phase give maximum root real parts -0.22457 and -0.08727: each alone is stable. These are numerical checks, not symbolic margins.

For stable S, the exact geometric identity is

$$\|S\|_\infty = \frac{1}{\inf_{\omega \in \mathbb{R} \cup \{\infty\}} |1 + L(i\omega)|}. \tag{12}$$

This measures absolute loop displacement from -1, not distance from the origin. $\|T\|_\infty$ instead measures amplification of **relative** loop error, as we now derive.

5 Specify a set of possible plants, not a vague percentage

Suppose the real plant is H. At a frequency where $G \neq 0$, write the relative error

$$\frac{H-G}{G} = \frac{H}{G} - 1. \tag{13}$$

A stable, proper nonzero weight W assigns a frequency-dependent allowance. A high-pass weight represents good low-frequency identification and poorer high-frequency knowledge. An uncertainty weight has units of relative error, while a performance weight is a reciprocal allowance for a specified signal.

There are two useful uncertainty classes; they must not be silently identified.

The book's class (Definition 2.11) consists of proper rational H with the same number of open-right-half-plane poles as G, identical imaginary-axis poles including multiplicity, and

$$\left|\frac{H(i\omega)}{G(i\omega)} - 1\right| < |W(i\omega)| \tag{14}$$

where $W \neq 0$, including the limiting value at infinity. At zeros of W use the matching/limiting convention $\frac{H}{G} = 1$; ratios at poles or zeros of G are interpreted by removable limits. These are **strict** frequency inequalities. Pole-count restrictions are restrictions on H, not "unstable poles of a stable perturbation."

The stable-block class starts from $H = G(1 + W\Delta)$ with $\Delta \in \mathcal{RH}_\infty$ and $\|\Delta\|_\infty < 1$. Here $\mathcal{RH}_\infty$ means real-rational proper stable functions. It is often a convenient subset of the book's model, but admissibility also requires that $1 + W\Delta$ not cancel any unstable or imaginary-axis poles of G.

The distinction is substantive: let

$$G = \frac{1}{s-1}, \quad H = \frac{1}{s-2}, \quad W = \frac{1.1}{s+2}. \tag{15}$$

Both plants have one unstable pole and meet the frequency bound, since

$$\Delta = \frac{\frac{H}{G} - 1}{W} = \frac{s+2}{1.1(s-2)}, \quad \|\Delta(i\mathbb{R})\|_\infty = \frac{1}{1.1} < 1. \tag{16}$$

This Δ is **unstable**. It belongs to a frequency description of the book's class, not to a stable uncertainty block. With $K = 3$, both endpoint loops are stable, yet the intermediate plant $(1-t)G + tH$ generally has **two** unstable poles. An arbitrary straight-line homotopy of plants need not stay in the prescribed pole-count class.

In practice, first validate that the physical parameter family is covered. The resulting disk model normally contains many additional dynamic plants. A test exact for that larger class may be conservative for the original parameter family. This is a modeling cost, not a defect in the theorem.

> Always state: stable block or pole-count model? Strict or closed bound? Rational dynamics or delays? Each choice changes what an endpoint or a failed test means.

6 Derive weighted small gain from the feedback denominator

Assume K internally stabilizes nominal G, W is stable proper, and the interconnections are well posed. For $H = G(1 + W\Delta)$,

$$1 + HK = (1 + L)(1 + WT\Delta), \quad S_H = (1 + HK)^{-1} = \frac{S}{1 + WT\Delta}. \tag{17}$$

The uncertain feedback is therefore a second loop: the stable nominal map $M = WT$ feeds the uncertainty block. If Δ is stable and $\|M\|_\infty \|\Delta\|_\infty < 1$, then $1 + M\Delta$ cannot vanish in the closed right half-plane. Indeed the maximum modulus theorem extends the boundary gain bound into the right half-plane, and $|M\Delta| < 1$ excludes the value -1. The same strict inequality at infinity prevents algebraic ill-posedness.

For **every stable block with norm strictly below one**, the exact scalar test is

$$\|WT\|_\infty \le 1. \tag{18}$$

Necessity has a useful mechanism. If $|M(i\omega_0)| > 1$, choose $\delta_0 = -\frac{1}{M(i\omega_0)}$, with $|\delta_0| < 1$. A stable real all-pass factor times a real gain can interpolate this complex value at the conjugate frequencies $\pm i\omega_0$ (exercise 5). Then $1 + M\Delta = 0$ at that frequency. There is an admissible perturbation causing a boundary pole, not merely a large nominal response.

For the book's larger pole-count class, Theorem 2.13 gives the same exact test, under nominal internal stability. Its sufficiency uses Nyquist and fixed endpoint pole counts, not the assertion that the normalized Δ is stable. Necessity additionally avoids cancelling plant poles by a narrow stable interpolation filter (book pp. 30–33). These are important qualifications to the elementary stable-block proof above.

6.1 Equality is a convention, not a numerical tolerance

For a **closed** stable-block ball $\|\Delta\|_\infty \le 1$, the safe general criterion is $\|M\|_\infty < 1$, not ≤ 1. Example: $G = 1, K = 1, W = 2$ gives $T = \frac{1}{2}$, $M = 1$. Every strict-block perturbation is safe, but $\Delta = -1$ makes $H = -1$ and $1 + HK = 0$. Equality of the product bound can therefore destroy well-posedness. Special closed-ball cases can survive equality (exercise 6); they need a separate proof.

With an open uncertainty allowance $r|W|$, the largest robust radius is $r_* = \frac{1}{\|WT\|_\infty}$. At $r = r_*$ every **strictly interior** model is stable, but the family need not have a uniform stability margin. For closed balls the same number is generally a supremum, not an included endpoint.

Geometrically Equation 18 says $|WL| \le |1 + L|$ at every frequency: the open disk of possible loop values centered at L cannot contain -1. This is precisely the mixed gain/phase protection that separate classical margins do not provide.

7 Performance spends part of the same distance budget

A stable proper weight W_p expresses the desired error envelope:

$$\left\|W_p S\right\|_\infty \le 1 \quad \Longleftrightarrow \quad |S(i\omega)| \le \frac{1}{|W_p(i\omega)|}. \tag{19}$$

Large low-frequency $|W_p|$ demands accurate tracking; a small high-frequency weight allows error where measurement noise or uncertainty makes feedback unwise. This is an energy/sinusoidal-gain specification, not a guarantee on step overshoot or saturation.

At a fixed frequency let $a = W_p S$, $b = WT$. From Equation 17,

$$|W_p S_H| = \frac{|a|}{|1 + b\delta|}. \tag{20}$$

When $|b| < 1$, the infimum of $|1 + b\delta|$ over $|\delta| < 1$ is $1 - |b|$: choose δ with phase opposing b, and let its magnitude tend to one. Hence robust stability **and** robust performance for the book's strict class are equivalent to

$$|W_p S| + |WT| \le 1 \quad \text{at every frequency, including infinity.} \tag{21}$$

This is Theorem 2.21. The boundary case $|b| = 1$ forces $a = 0$ under this test; handle zero weights by limits. A single constant complex δ is only frequency geometry; the all-pass interpolation argument supplies an actual real dynamic perturbation when the inequality is violated. For a closed uncertainty ball, use strict margin for stability.

The geometry is now $|W_p| + |WL| \le |1 + L|$. Performance reserves a disk around -1; uncertainty reserves a disk around L. They must not overlap. Checking nominal performance and robust stability separately is necessary but not sufficient: $0.7 + 0.6 > 1$.

7.1 A thermal loop: stable uncertainty need not preserve the error target

Normalize time and temperature so $\dot{x} = -x + u$, $y = x$, and choose $K = 3$. Then $S = \frac{s+1}{s+4}$, $T = \frac{3}{s+4}$. Take $W = 0.2$, a 20% dynamic actuator-error allowance (including static gain changes, but also phase changes). Robust stability is easy: $\|WT\|_\infty = 0.15$. But for $W_p = 1$,

$$|S| + |WT| = \frac{\sqrt{\omega^2 + 1} + 0.6}{\sqrt{\omega^2 + 16}}. \tag{22}$$

Put $t = \sqrt{\omega^2 + 1} \ge 1$. Differentiating $\frac{t+0.6}{\sqrt{t^2+15}}$ gives its maximum at $t = 25$, namely $\sqrt{1 + \frac{0.6^2}{15}} = 1.011929$ at $\omega = \sqrt{624} = 24.9800$. Thus robust performance fails even though nominal performance has norm one and robust stability has ample margin. The uncertainty can rotate phase as well as change gain.

If $W_p = 0.8$ instead, Cauchy–Schwarz gives the **continuous** certificate

$$\frac{0.8t + 0.6}{\sqrt{t^2 + 15}} \le \sqrt{0.8^2 + \frac{0.6^2}{15}} = 0.814862 < 1. \tag{23}$$

This relaxed specification is robustly feasible. Conversely a failed sufficient bound would not prove infeasibility: for example $|a| + |b| \le \sqrt{2}\sqrt{|a|^2 + |b|^2}$ is an upper bound, not an equality. Failure of its stacked-norm test only says "undecided."

7.2 Read the curves, then decide what they establish

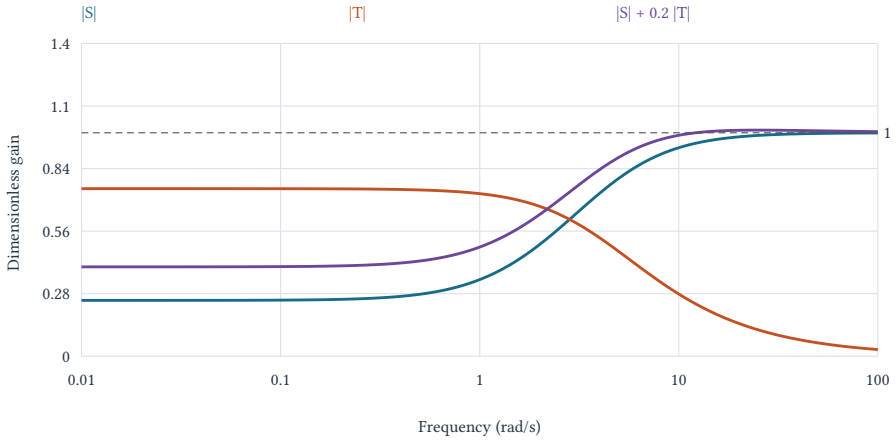


Figure 2: Thermal loop. Tracking improves at low frequency, while T transmits low-frequency reference and noise. The purple robust-performance budget slightly exceeds one at high frequency.

A plot samples frequencies. Even a million samples can miss a narrow resonance or a tail maximum. For a stable scalar rational transfer F, a continuous norm computation can use a Hamiltonian-based algorithm, or solve for all stationary points of $|F(i\omega)|^2$ as a rational function of ω^2, and compare $\omega = 0, \infty$. Such methods still use floating-point tolerances; they are not interval-arithmetic proofs.

The sum in Equation 21 is not itself the magnitude of one scalar rational transfer. Our satellite script uses a dense logarithmic search plus local maximization; it supplies strong numerical evidence, not a continuous certificate. The thermal example's analytic maximum is stronger. A sampled value **above** one, by contrast, is already a valid counterexample frequency (allowing for numerical error); no missing peak can undo it.

8 A delayed flexible satellite: three distinct boundaries

For exercises 4–5 use the actual printed plant, not its ten-times-more-damped worked example:

$$G = 0.036\frac{s + 25.28}{s^2(s^2 + 0.00396s + 1)}, \tag{24}$$

$$K = \frac{7.9212(s + 0.1818)(s^2 - 0.2244s + 0.8981)}{(s^2 + 3.899s + 4.745)(s^2 + 1.039s + 3.395)}. \tag{25}$$

There is an attitude double integrator and a lightly damped flexible mode; the dynamics are not a symmetric single resonance. The nominal unreduced closed-loop poles have maximum real part -0.2070. A measurement delay τ changes the loop to $e^{-s\tau}L$. For stability this is equivalent to putting the delay in the plant; the physical reference/output maps differ if one also changes where the reference is delayed.

For all delays $0 \le \tau \le h$, use the first-order cover

$$W_h = s\frac{h}{1 + s\frac{h}{\sqrt{12}}}, \quad |e^{-i\omega\tau} - 1| < |W_h(i\omega)| \quad (\omega \neq 0). \tag{26}$$

It matches the first-order low-frequency error and overbounds the oscillatory high-frequency error. A pure delay is not rational and has no single value at infinity; the exercise explicitly permits the sufficient delay extension. The rational model-set theorem is not being applied literally to it. Exercise 4 below verifies the cover and its endpoint issue.

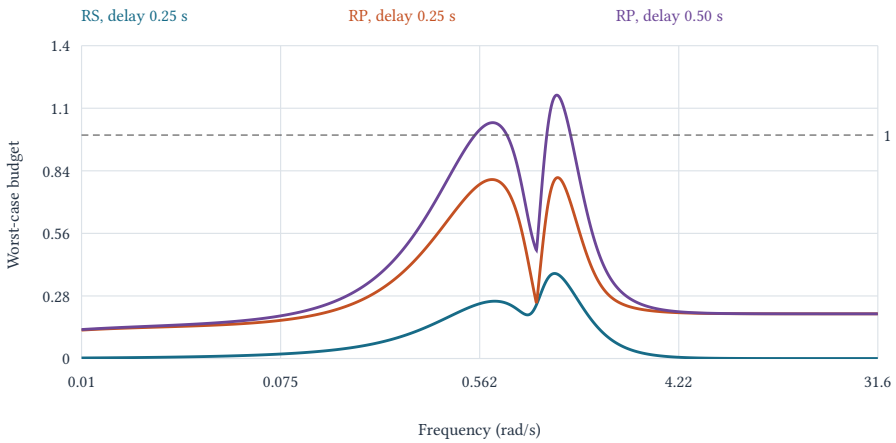


Figure 3: Same satellite and controller. Blue spends only the stability budget. Orange and purple add weighted tracking error. A 0.50 s uncertainty cover violates performance before it violates robust stability.

Use the book's performance weight

$$W_p = \left(\frac{\frac{s}{\sqrt{5}} + \omega_B}{s + \omega_B \sqrt{10^{-3}}} \right)^2, \quad \omega_B = 0.1. \tag{27}$$

At zero its magnitude is 1000, so the allowed error is 0.001; at infinity its magnitude is $\frac{1}{5}$, allowing sensitivity up to 5. Its bandwidth parameter is a specification parameter, not automatically the physical closed-loop −3 dB bandwidth.

Question	Computed answer
Nominal $\|S\|_\infty$, $\|T\|_\infty$	2.4331, 1.8286
$\|W_{0.25}T\|_\infty$; reciprocal uncertainty radius	0.3805; 2.6278
Largest h passing covering-class RS test	0.6720 s
First actual pure-delay crossing	0.7669 s
RP budget at $h = 0.25$	0.8095
Largest h passing covering-class RP test	0.3781 s
Largest ω_B at $h = 0.25$ passing RP	0.20066 rad/s

The rational stability norm is also checked by SLICOT at relative tolerance 10^{-9}. At $h = 0.25$, its stacked norm gives the continuous RP upper bound $\sqrt{2}\left\| \begin{pmatrix} W_p S \\ W_h T \end{pmatrix} \right\|_\infty = 0.851515 < 1$. The sharper RP maxima and thresholds are grid-plus-refinement estimates. Polynomial root enumeration checks all three unit-gain crossings; their delay residuals are below 10^{-12} and the first moves rightward. No nominal unstable cancellation occurs.

> 0.6720 s is a sufficient delay guarantee obtained from a larger uncertainty set, not the actual delay margin. Failure of that cover test above 0.6720 s does not prove that the physical delay is unstable. Likewise 0.3781 s is not a demonstrated physical-delay performance boundary.

For the stated 0.25 s delay, both requirements pass with useful numerical margin. Retain the controller for these linear specifications, subject to actuator and model validation. If the uncertainty allowance grows, either relax the performance requirement or redesign; do not extrapolate a gain margin into a tracking guarantee.

9 Exercise guide: all ten, with the technical gaps identified

The following follows the actual exercise statements on pp. 42–45. Worked arguments are given where short; the general pole-count homotopy in exercise 9 is labelled an advanced outline. This guide is separate from the main reading, not a prerequisite for it.

9.1 Exercise 1 — compute before cancelling

The book's system matrices mean $A_G = \left(\begin{smallmatrix} 0 & 1 \\ -2 & -3 \end{smallmatrix}\right)$, $B_G = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$, $C_G = (\alpha \ \ -1)$, $D_G = 0$; the controller has $A_K = B_K = C_K = 1$, $D_K = 0$. For the product driven by e, with state ordered (x_G, x_K),

$$A_L = \begin{pmatrix} 0 & 1 & 0 \\ -2 & -3 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad B_L = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad C_L = (\alpha \ \ -1 \ \ 0), \quad D_L = 0. \tag{28}$$

The inverse of $1 + L$ has realization

$$A_S = A_L - B_L C_L = \begin{pmatrix} 0 & 1 & 0 \\ -2 & -3 & 1 \\ -\alpha & 1 & 1 \end{pmatrix}, \tag{29}$$

$$B_S = B_L, \quad C_S = -C_L, \quad D_S = 1. \tag{30}$$

Its state matrix is also the physical closed-loop matrix. Direct elimination gives

$$L = \frac{\alpha - s}{(s+1)(s+2)(s-1)}, \tag{31}$$

$$p = s^3 + 2s^2 - 2s + \alpha - 2, \quad S = \frac{(s+1)(s+2)(s-1)}{p}. \tag{32}$$

At $\alpha = 1$, $p = (s-1)(s^2 + 3s + 1)$. The state eigenvalues are $1, -0.381966, -2.618034$, so this realization is not Hurwitz. Reduced transfers are

$$S = \frac{(s+1)(s+2)}{s^2 + 3s + 1}, \quad GS = -\frac{1}{s^2 + 3s + 1}, \tag{33}$$

$$KS = \frac{(s+1)(s+2)}{(s-1)(s^2 + 3s + 1)}. \tag{34}$$

Only the last exposes the unstable mode. With $\alpha = 1.001$ or 0.999, the unstable root becomes 0.9998 or 1.0002 and the cancellation in S disappears. In fact implicit differentiation at $(s, \alpha) = (1, 1)$ gives $\frac{\mathrm{d}s}{\mathrm{d}\alpha} = -\frac{1}{5}$, explaining the numbers. The instability was present even before perturbation. `scripts/ex1.m` checks the unreduced eigenvalues and the reduced channel denominators.

9.2 Exercise 2 — choose the disturbance location correctly

(a) The signal equations Equation 1 with $n = 0$ give Equation 5, including the minus sign in $-GS$. Putting d at the **output** instead would give a different transfer matrix; it would not be Figure 16's experiment. Multiplication verifies the inverse because $S + GKS = 1$ and the off-diagonal products cancel.

(b) The plant/controller pair of section 2 has stable S and KS but unstable GS. It is the required example with only entry $(1, 2)$ unstable. **(c)** If G is stable and KS is stable, then $S = 1 - G(KS)$ and GS are stable. Conversely internal stability includes KS. Thus for stable G, one channel suffices; the argument depends on stability of G, not merely on properness or nominal output tracking.

9.3 Exercise 3 — signs and contour detours

Apply the argument principle to $f = 1 + L$ on the clockwise contour of section 3. It gives Equation 9, $N = P - Z$, with $P = n_{0+}$ when all axis poles are included by leftward detours. There are no boundary zeros because the image avoids -1. For sufficiently small detours any axis zeros would lie on the enclosed straight-axis part, and would violate that condition; the limit contour therefore checks them too. At infinity $L(\infty) \neq -1$ ensures a proper inverse. Hence stability is exactly $Z = 0$, or $N = n_{0+}$. Avoiding -1 **alone** does not imply $Z = 0$.

9.4 Exercise 4 — derive the delay cover, then interpret the margin

(a) In the family $W_h = s\frac{h}{1+s\frac{h}{c}}$, the low-frequency slope is fixed to match the delay. Put $x = |\omega|h$. The delay magnitude is $2|\sin(\frac{x}{2})|$, not a function converging to 2 at infinity. The weight magnitude is $\frac{x}{\sqrt{1+\frac{x^2}{c^2}}}$. Expansion gives $2\sin(\frac{x}{2}) = x - \frac{x^3}{24} + O(x^5)$ and $|W_h| = x - \frac{x^3}{2c^2} + O(x^5)$, so coverage near zero requires $c^2 \geq 12$.

For $c^2 = 12$ the global strict inequality also holds. If $x \geq \sqrt{6}$, the weight is at least 2, and strictness follows (at $x = \sqrt{6}$ the sine is strictly below one). For $0 < x \leq \sqrt{6}$ put $y = \frac{x}{2}$, $t = y^2 \leq \frac{3}{2}$. The alternating cosine series gives

$$\sin^2 y \leq t - \frac{t^2}{3} + 2\frac{t^3}{45} < \frac{t}{1+\frac{t}{3}}, \tag{35}$$

since the difference in the second inequality is $t^3\left(\frac{1}{9(1+\frac{t}{3})} - \frac{2}{45}\right) > 0$. Taking square roots proves the cover. It is tight **within this fixed-slope first-order family**, not a unique best weight over all dynamical orders. Since $|W_h(i\omega)|$ increases with h, it also covers every smaller delay.

(b) Multiplication by $e^{-s\tau}$ adds no finite poles or zeros, so the satellite's double pole at zero and open-right-half-plane pole count are unchanged. At zero, the relative error and W_h both vanish. However $\lim_{s\to 0} \frac{e^{-sh}-1}{W_h(s)} = -1$. Thus the normalized block lies on the **closed** unit-ball boundary, not in the strict stable rational ball. Moreover delay is irrational. The exercise's extension is needed. One safe justification for a strict passing test is to enlarge h slightly while retaining $\|W_h T\| < 1$; the enlarged cover makes the normalized zero-frequency value strictly smaller than one. More directly the strict product small-gain theorem extends to bounded analytic causal delay blocks. An exact equality threshold requires additional limiting analysis and is not being certified here for the delay family.

(c–d) Section 8 answers the numerical questions. The printed exercise's expression $\|W_h S\|$ omits the loop factor; Theorem 2.13 requires $W_h T$, not $W_h S$. The radius 2.6278 scales the entire weight, not time itself. The maximum delay passing that nonlinear weight test is about 0.6720 s; actual first crossing is 0.7669 s. For 0.25 s keep the controller on the stated linear evidence, but also check performance, actuator authority, and that the identified model is covered.

9.5 Exercise 5 — produce a bad plant, not just a failed bound

(a–c) With W_p as in section 8, maximize $q_{h(\omega)} = |W_p S| + |W_h T|$. The refined search returns 0.8095 at $\omega = 1.2383$ for $h = 0.25$, and the thresholds $h \approx 0.3781$ s and $\omega_B \approx 0.20066$ rad/s. All are for damping 0.00396, not 0.0396. The index increases pointwise with h and with ω_B for this weight, justifying the scalar threshold searches. The script's local refinement does not prove every maximum was found; zero and infinity limits and a broad grid must still be considered.

(d) Choose $h = 0.5$. At the violating frequency $\omega_0 = 1.2274$, set $u_0 = 1 + L(i\omega_0)$ and $v_0 = W_h(i\omega_0)L(i\omega_0)$. Numerically $|W_p| = 0.2066$, $|u_0| = 0.4805$, $|v_0| = 0.3597$. The phase choice

$$\delta_0 = 0.999 \exp(i(\arg u_0 - \arg v_0 + \pi)) \tag{36}$$

opposes v_0 to u_0 and gives $|u_0 + \delta_0 v_0| = 0.1212 < |W_p|$. To realize it with real dynamics choose $\alpha = \pm\,|\delta_0|$ and

$$\beta = i\omega_0 \frac{\alpha - \delta_0}{\alpha + \delta_0}, \quad \Delta(s) = \alpha \frac{s - \beta}{s + \beta}. \tag{37}$$

The ratio makes β real; choose the sign giving $\beta > 0$. The resulting stable all-pass times a gain has constant boundary magnitude $0.999 < 1$. Here it is

$$\Delta(s) \approx -0.9990 \frac{s - 0.6601}{s + 0.6601}. \tag{38}$$

Then $H = G(1 + W_h\Delta)$ is in the rational uncertainty class: it has no new unstable poles, and $1 + W_h(0)\Delta(0) = 1$ preserves the satellite's double integrator. Its closed-loop polynomial has maximum root real part -0.0361; a frequency search gives $\|W_p S_H\|_\infty \approx 1.7057$. Already the value at ω_0 exceeds one, so this is a constructive performance counterexample without needing a norm certificate. It is **not** proof that a physical delay of 0.5 s violates performance: this H is a different dynamic plant in the larger cover class.

9.6 Exercise 6 — a closed-ball endpoint that really is safe

(a) $G_0 = \frac{1}{s}$, $K = 1$ gives $T = \frac{1}{s+1}$ and $\|T\|_\infty = 1$. **(b)** For real stable rational Δ with norm at most one and $\Delta(0) = -1$, write $\Delta = -1 + a_1 s + O(s^2)$. The maximum modulus theorem gives $|\Delta(h)| \le 1$ for small real $h > 0$; expanding shows $a_1 \ge 0$. Equivalently $\frac{1+\Delta}{1-\Delta}$ has nonnegative real part and derivative $\frac{a_1}{2}$ at zero. The constant case $\Delta = -1$ satisfies this too.

(c) $S_\Delta = \frac{s}{s+1+\Delta(s)}$. In the closed right half-plane, a denominator zero requires $|\Delta(s)| = |1 + s| \le 1$, which forces $s = 0$. If $\Delta(0) \neq -1$, even that zero is absent. Otherwise the derivative of the denominator at zero is $1 + a_1 \ge 1$, so the zero is **simple** and is removed by the numerator. The inverse is proper because Δ is proper and s dominates at infinity. Hence S_Δ is stable for the entire closed ball. This special cancellation concerns a changing plant pole; it is not a license to cancel an unstable mode in a fixed physical realization.

9.7 Exercise 7 — internal Nyquist uses the uncancelled pole count

(a) The identity Equation 8 proves the equivalence. If S is stable but p has a root λ in the closed right half-plane, then $D_{G(\lambda)}D_{K(\lambda)} = 0$; also $N_{G(\lambda)}N_{K(\lambda)} = 0$. Coprimeness within each factor forces a cross cancellation: either $N_G = D_K = 0$ or $N_K = D_G = 0$ at that point. With no such cancellation, no bad root of p can exist. Conversely, if a cross cancellation occurs, say $N_{G(\lambda)} = D_{K(\lambda)} = 0$, then $D_{G(\lambda)}N_{K(\lambda)} \neq 0$. Entry KS in Equation 8 has a pole there, so internal stability fails.

(b) Let m count all closed-right-half-plane poles of G and K, before multiplying, and let P count poles of the reduced L. Then $P \leq m$. Nyquist always gives $N = P - Z \leq P \leq m$. Therefore $N = m$ forces **both** $Z = 0$ and $P = m$, which excludes unstable cancellations. This proves the converse as well as the forward implication; one cannot simply assume the equality of the two pole counts when proving it. As always, exclude the point -1 on the complete contour and require well-posedness at infinity. Those qualifications are necessary even if omitted from a short exercise statement.

9.8 Exercise 8 — why boundary bounds control the whole half-plane

(a) A continuous subharmonic f satisfies the circle mean inequality. On any compact set K, perturb it to $f_{\varepsilon(z)} = f(z) + \varepsilon\ |z|^2$. If this attained its maximum at an interior point z, choose a small circle inside K and inside the open right half-plane. The mean of $|z + re^{it}|^2$ is $|z|^2 + r^2$. Thus the subharmonic inequality for f implies

$$f_{\varepsilon(z)} \leq \frac{1}{2\pi}\int_0^{2\pi} f_{\varepsilon(z+re^{it})}\,\mathrm{d}t - \varepsilon r^2, \tag{39}$$

contradicting maximality. Its maximum is therefore on the boundary. Let ε decrease to zero to get $\sup_K f = \sup_{\partial K} f$. Points of K on the imaginary axis are boundary points automatically; disconnected sets cause no difficulty.

(b) Cauchy–Schwarz gives $|x^*My| \leq \|My\| \leq \|M\|$ for unit vectors. Choose a maximizing right singular vector y and $x = M\frac{y}{\|My\|}$ (the zero matrix is trivial) to attain equality. Thus $\|M\| = \max|x^*My|$.

(c) At a chosen center z, fix maximizing unit vectors x, y. Analyticity of $g(w) = x^*G(w)y$ gives the scalar mean inequality for $|g|$. On the surrounding circle, $|g(w)| \leq \|G(w)\|$, so

$$\|G(z)\| \leq \frac{1}{2\pi}\int_0^{2\pi} \left\|G(z + re^{it})\right\|\,\mathrm{d}t. \tag{40}$$

This directly proves subharmonicity without invoking an unjustified supremum rule.

(d) Apply (a) to closed right half-disks. A proper rational stable matrix tends uniformly to $G(\infty)$ on the large semicircle; its norm is already a limit of imaginary-axis values. Sending the radius to infinity proves

$$\sup_{\Re z \geq 0}\|G(z)\| = \sup_{\omega\in\mathbb{R}}\|G(i\omega)\| = \|G\|_\infty. \tag{41}$$

This is the analytic fact behind small gain, not evidence supplied by a Bode grid.

9.9 Exercise 9 — pole-count preservation (advanced proof outline)

Here Δ has no imaginary-axis poles and $|\Delta(i\omega)| < 1$, including infinity, but is **not assumed stable**. Keep this distinction: the exercise addresses exactly the case where the elementary stable-block proof is insufficient. Put $T = \frac{L}{1+L}$, $F = 1 + TW\Delta$ and $L_\Delta = L(1 + W\Delta)$.

(a) Since T, W, Δ are analytic on the axis and proper, F has no poles there or at infinity. The strict boundary inequality $|TW\Delta| < 1$ gives $F \neq 0$ there, so its inverse has no poles there either. This says nothing yet about right-half-plane zeros or poles of F.

(b) At an axis pole λ of L, analytically extended $T(\lambda) = 1$. Therefore $1 + W(\lambda)\Delta(\lambda) = F(\lambda) \neq 0$, and that pole survives with its full multiplicity. Away from those poles, multiplying by an axis-analytic factor creates no new axis poles. We do **not** need $1 + W\Delta$ to be nonzero everywhere on the axis; it may vanish at points where L is finite.

(c) Use the identity $1 + L_\Delta = (1 + L)F$. The strict boundary bound makes the image of F lie in the open disk centered at 1 with radius 1 on the limiting Nyquist boundary, so its winding about zero is zero. On sufficiently large outer arcs and sufficiently small detours the strict inequality persists by properness and continuity, even though F may have poles **inside**. The meromorphic argument principle then gives equality of the differences "zeros minus poles" for $1 + L_\Delta$ and $1 + L$. Now $S = 1 - T$ is stable by hypothesis, and S_Δ is stable by the additional assumption in (c), so neither denominator function has zeros in the open right half-plane. Their pole counts there must consequently agree. Adding 1 does not alter finite poles, hence L and L_Δ have the same open-right-half-plane pole count.

This outlines the contour argument, with the detour construction as in book pp. 30–31. It does not replace a full contour parametrization proof. In particular one must not infer that F is stable from its boundary values: unstable poles of Δ can persist.

A useful warning from the model set.

The plants $\frac{1}{s-1}$ and $\frac{1}{s-2}$ in section 5 show why this exercise is needed. With $K = 3$, their sensitivities are $\frac{s-1}{s+2}$ and $\frac{s-2}{s+1}$, both stable. But $(1-t)G + tH$ for $0 < t < 1$ has denominator $(s-1)(s-2)$ and numerator $s - 2 + t$. Its closed-loop characteristic polynomial is $s^2 - 4 + 3t$, with a positive root $\sqrt{4 - 3t}$. Thus every intermediate model is unstable in feedback, despite stable endpoints and despite a small boundary relative-error plot.

The failure is the extra unstable plant pole in the intermediate models, not a paradox in small gain. Fixing the pole count is essential when frequency error bounds are used without a stable-block assumption. This is why the book's proof separates the boundary homotopy argument from its final endpoint pole-count argument.

9.10 Exercise 10 — delays add infinitely many roots, but not arbitrarily

Let proper $L_0 = \frac{N}{D}$ be in coprime form and $\tau > 0$. Write

$$L_\tau = L_0 e^{-s\tau}, \quad p_\tau = D + N e^{-s\tau}. \tag{42}$$

p_τ is an entire **quasipolynomial**, not in general a polynomial.

(a) $S_\tau = \frac{D}{p_\tau}$. If $D(\lambda) = p_{\tau(\lambda)} = 0$, then $N(\lambda)e^{-\lambda\tau} = 0$, impossible by coprimeness. So no finite zeros of p_τ cancel, and its closed-right-half-plane zeros are exactly the poles in question.

(b) Put $c = L_0(\infty)$ and suppose $|c| < 1$. Choose q with $|c| < q < 1$. For sufficiently large $|s|$ in the closed right half-plane, $|L_0(s)e^{-s\tau}| \le |L_0(s)| < q$, uniformly in $\tau \ge 0$. Thus $1 + L_\tau$ has no zeros outside a fixed disk. There are finitely many inside because p_τ is a nonzero entire function and zeros cannot accumulate in a compact set. This includes the equal-degree case $0 < |c| < 1$; strict properness is **not** implied by $|c| < 1$.

(c) Outside that disk, $|S_\tau| \le \frac{1}{1-q}$. Inside, $\frac{D}{p_\tau}$ is continuous and bounded if p_τ has no closed-right-half-plane zeros. It is analytic in the open half-plane. Bound the **ratio**, rather than relying on a polynomial degree argument for $\frac{1}{p_\tau}$ alone.

(d) $M|1 - e^{-i\omega\tau}| \le M|\omega|\tau \le MR\tau$. Choose $\tau_0 = \frac{1}{2MR}$ to obtain a strict bound even on the closed interval $[0, \tau_0]$.

(e) By (b), all potentially bad roots remain in one fixed disk for every small delay. On its closed right-half-disk, $p_\tau \to p_0 = D + N$ uniformly as $\tau \to 0$. Since p_0 has no zeros there, its continuous modulus has a positive minimum. Uniform convergence makes p_τ nonzero there for sufficiently small delay. Combined with the tail exclusion this proves the claim. This is also the rigorous "no roots arriving from infinity" argument behind the finite-frequency Nyquist delay computation.

(f) If $|c| > 1$, the limiting high-frequency equation is $e^{s\tau} = -c$, with roots

$$s_k = \frac{\log(|c|) + i(\arg(-c) + 2\pi k)}{\tau}. \tag{43}$$

Every root has positive real part $\frac{\log(|c|)}{\tau}$. To make the approximation rigorous, take disjoint circles of fixed radius $\varepsilon < \min\left(\frac{\log(|c|)}{\tau}, \frac{\pi}{\tau}\right)$ around s_k. On each circle $|e^{s\tau} + c|$ has a positive minimum independent of k. Because $L_0(s) - c = O\left(\frac{1}{|s|}\right)$, it is smaller than that minimum for large $|k|$. Rouché's theorem gives a root of $e^{s\tau} + L_0(s)$ in every such circle. These are zeros of p_τ (away from the finitely many rational poles), all in the right half-plane. So in fact infinitely many unstable roots exist for every positive delay. This elementary asymptotic argument avoids an incomplete use of Picard: Picard alone does not locate roots in the desired half-plane. The borderline $|c| = 1$ is deliberately not decided.

10 A reproducible analysis, and what it does not certify

The design logic is now a single chain. Specify where disturbances enter. Test all internal channels before simplifying. Use Nyquist to understand stability changes. Choose an uncertainty set that actually covers the plant error. Then ask separately whether the entire set is stable and whether its error remains within the chosen envelope.

For this chapter's examples, run from the repository root:

```
octave --no-gui --quiet Control/robust-control-02-siso/scripts/ex1.m
octave --no-gui --quiet Control/robust-control-02-siso/scripts/satellite_delay.m
octave --no-gui --quiet Control/robust-control-02-siso/scripts/figures.m
typst compile --root . \
  Control/robust-control-02-siso/robust-control-02-siso.typ \
  Control/robust-control-02-siso/robust-control-02-siso.pdf
```

`ex1.m` checks the physical state matrix, not only a minimalized sensitivity. `satellite_delay.m` computes characteristic roots, algebraically checked unit-gain crossings, delay crossing residuals and directions, the rational stability norm with the installed Octave control/SLICOT algorithm, a continuous RP upper bound, the RP searches, and a bad plant. `figures.m` regenerates the three SVG plots from transfer coefficients and checks the thermal example's analytic maximum. No external figure or package install is needed.

Three levels of evidence should remain distinct:

1. **Analytic results:** the thermal maxima, the delay cover, the scalar disk geometry, and the worked complex-analysis arguments hold on continuous domains.
2. **Continuous rational numerical algorithm:** the SLICOT norm check treats all frequencies with a stated solver tolerance. It is stronger than a grid but still floating-point computation, not a machine-verified exact certificate.
3. **Grid and refinement:** the displayed plots, RP thresholds and fifth-order Nyquist distance locate and illustrate behavior. The sampled counterexample frequency proves violation with a large numerical gap; passing values are numerical evidence unless supported by an independent continuous bound.

The satellite's delay extension and exercise 9′s general contour proof are explicitly qualified. Actuator saturation, nonlinearities, sampled implementation, and uncertainty identification are not certified by these linear calculations. Nor does a failure for the dynamic disk class imply that every physical parameter or pure delay model fails.

10.1 Source and onward connection

The authoritative source consulted is Scherer, **Theory of Robust Control**, chapter 2, pp. 16–45 (PDF page equals printed page), [1]. In particular: pp. 17–20 supply the signals and contour convention; pp. 22–26 the classical margins; pp. 27–39 the model set and exact SISO tests; pp. 40–41 internal stability; pp. 42–45 all ten exercises. The examples, corrections, derivations, and numerical checks above were developed against those statements, not copied as unverified inherited answers.

The next chapter replaces this special scalar loop by a general interconnection. The scalar sum-of-moduli performance test does **not** carry over unchanged to MIMO; structure matters. What survives is the habit: name the internal signals, define the uncertainty, prove stability, and only then interpret a gain bound.

[1] C. W. Scherer, "Theory of Robust Control." [Online]. Available: https://www.imng.uni-stuttgart.de/mst/files/RC.pdf